

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

Prof. Dr. Rüdiger Quay,

ruediger.quay@inatech.uni-freiburg.de

19.04.2026

Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

Contents

1	Abstract	6
2	Nonlinearity and Modulation	10
2.1	Phase- and Amplitude-modulated Signal Flows	10
3	Design for Linearity	13
4	6G: 2030 Design for Linearity	13
4.1	The classical trade-offs	18
4.2	Hints: Factors of Influence to ACPR and PAE	19
4.3	Tasks and Exercises: Vodaphone Challenge	23

Introduction

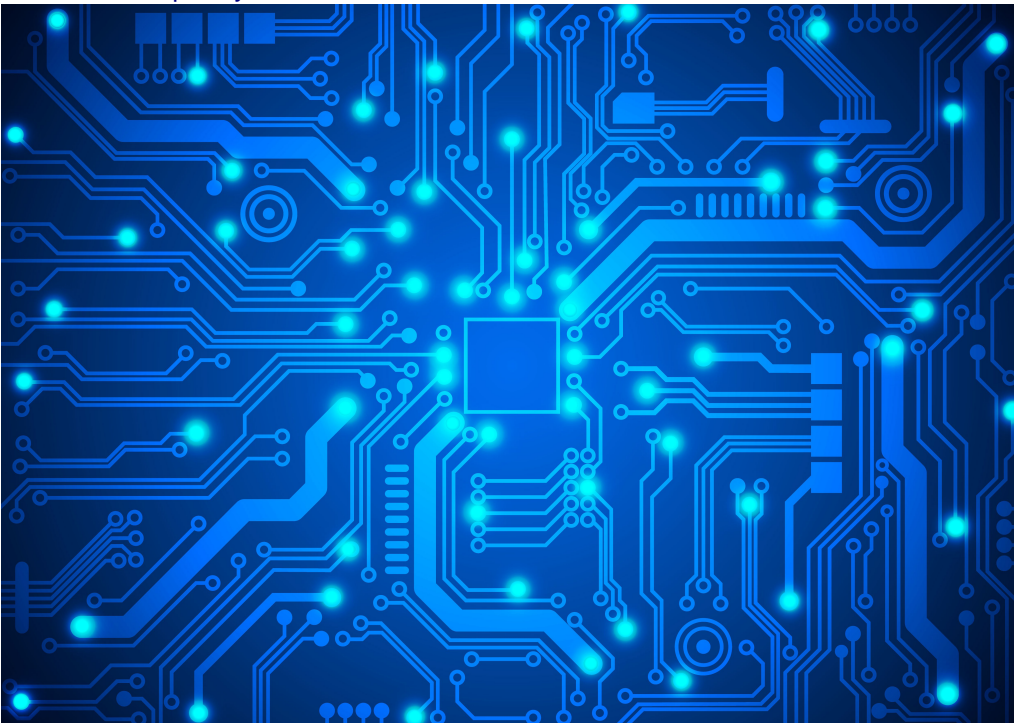


Introduction



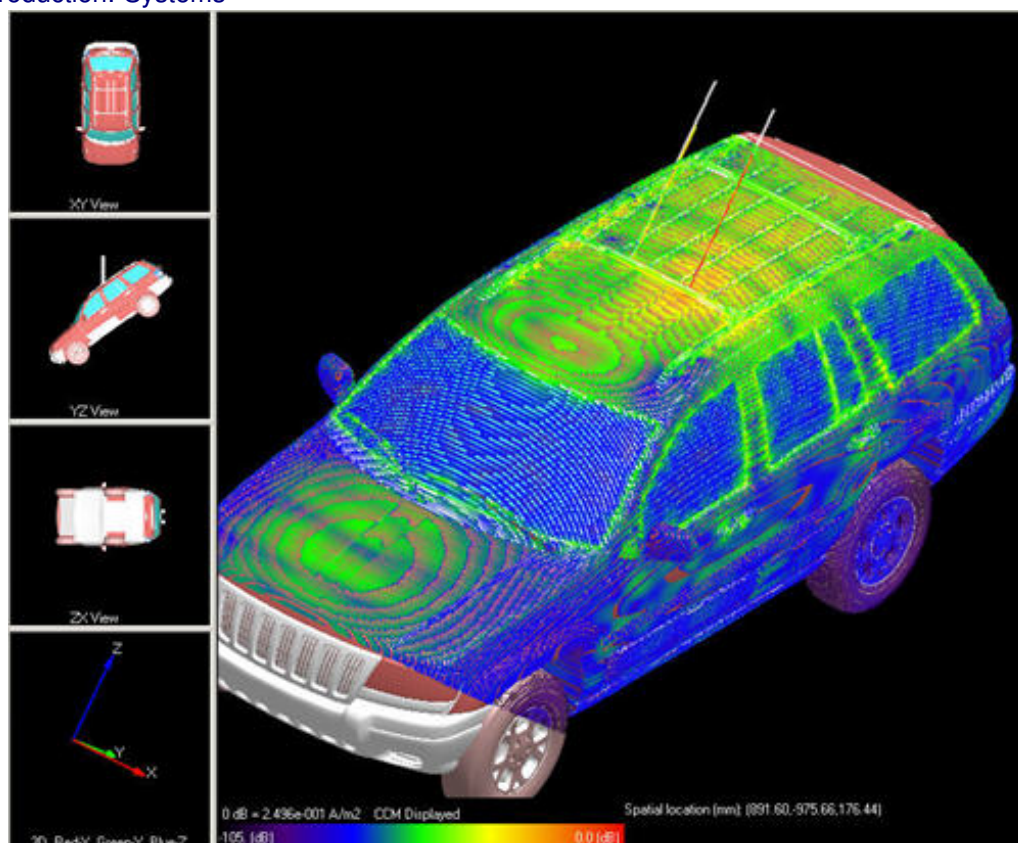
1.4

Introduction: Complexity



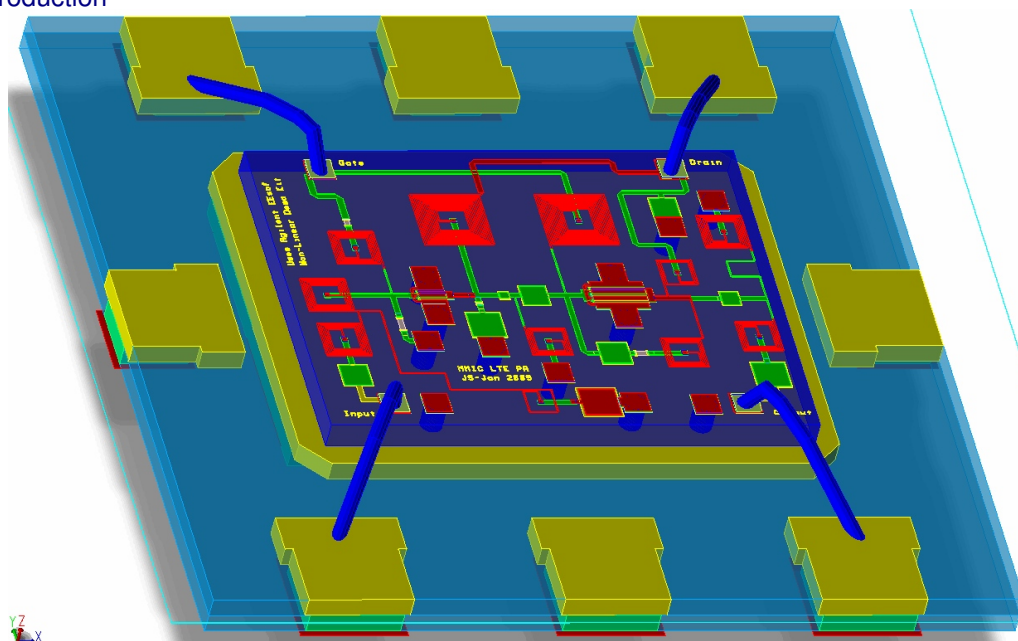
1.5

Introduction: Systems



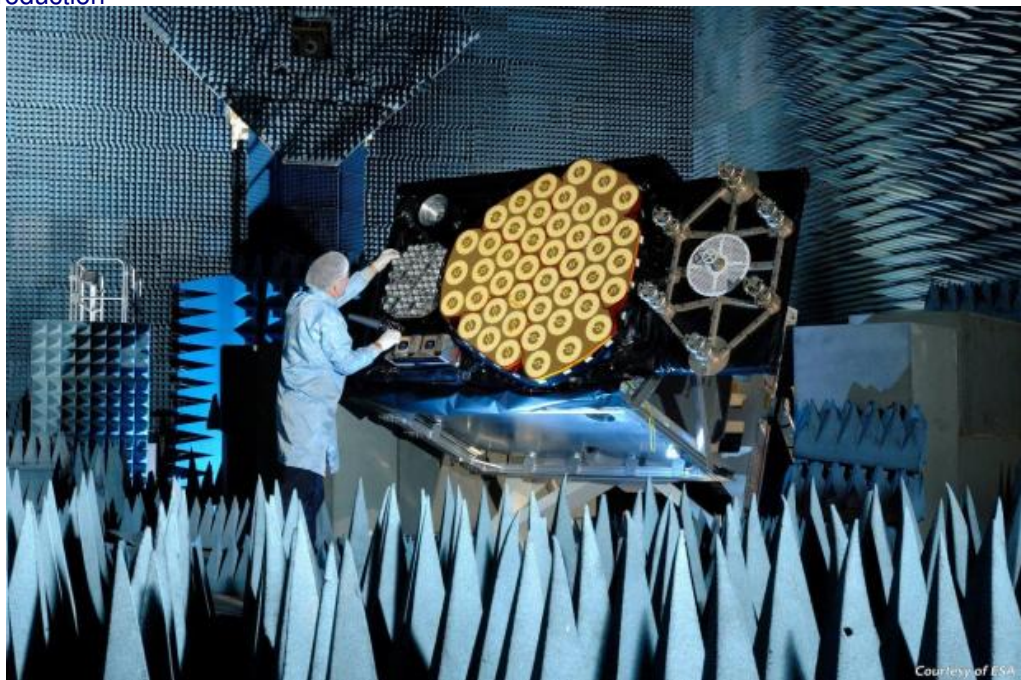
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

1.10

Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

1.13

Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

1.15

WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

1.17

Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

1.18

Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

1.19

Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: bash and tcsh

Start: ads after using ssh to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

1.20

Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

1.21

Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

Designing at Input and with more Complex Signals

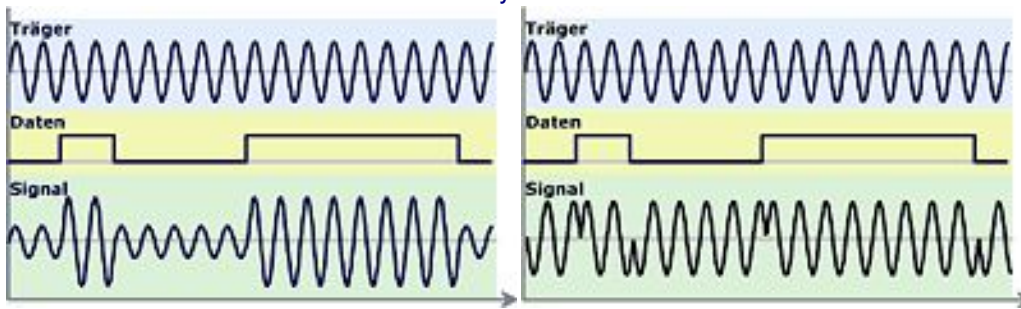
22. june 2026

9.24

2 Nonlinearity and Modulation

2.1 Phase- and Amplitude-modulated Signal Flows

Modulation: We do not look at sine waves only

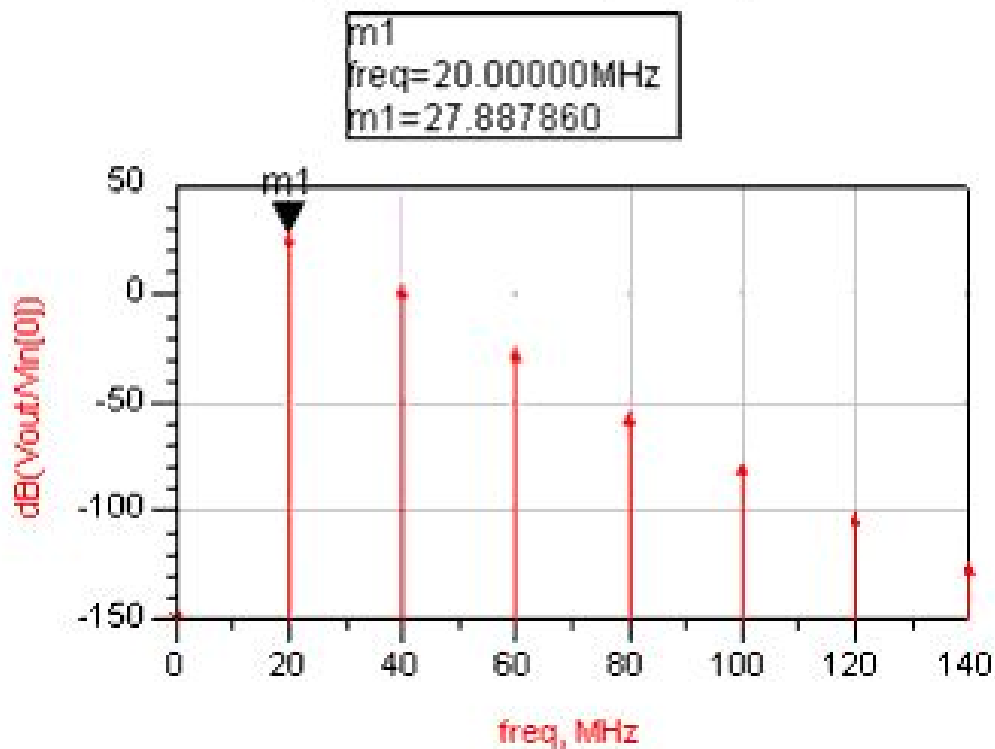


How to deal with this and which additional targets do we get from this ?

9.25

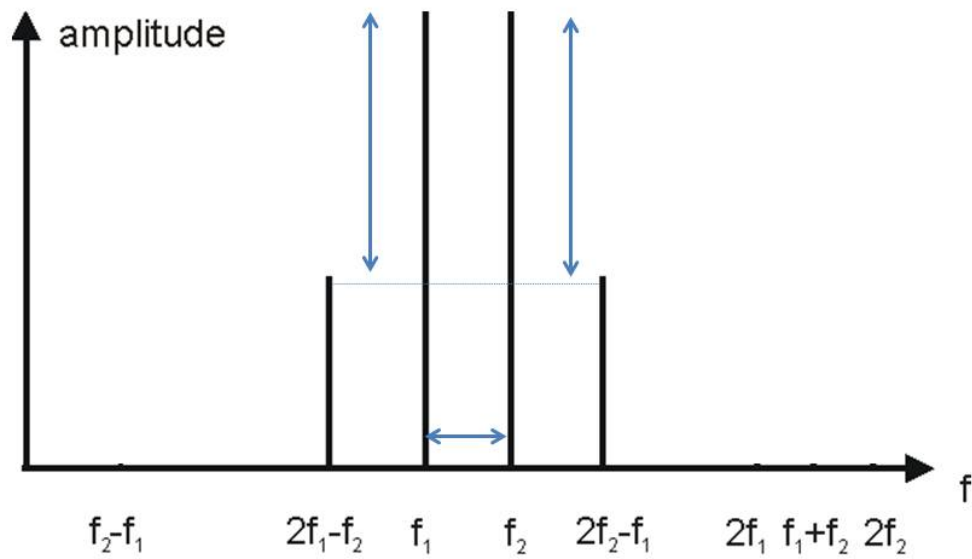
Nonlinearity: Harmonics

Fundamental and Harmonic Output
Voltages Relative to Input Voltage



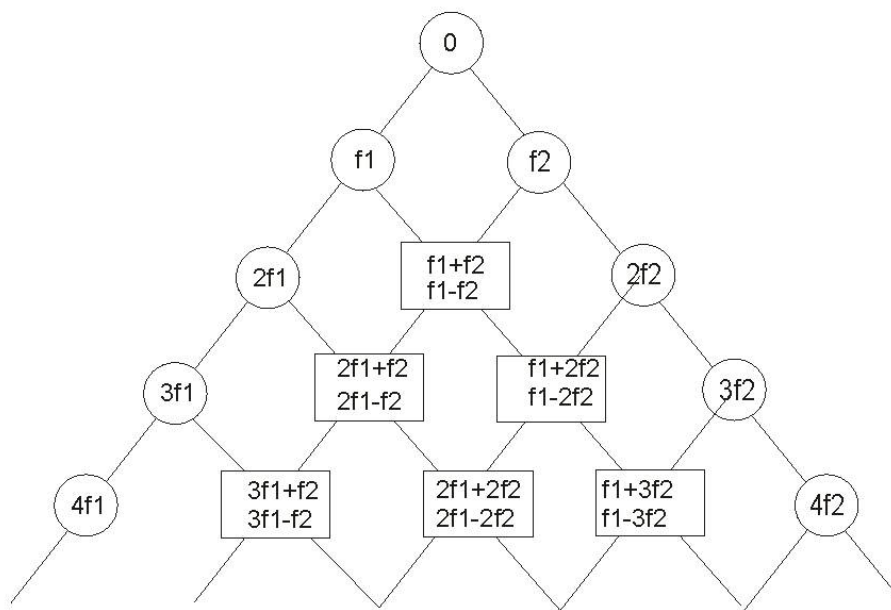
9.26

Nonlinearity: Intermodulation



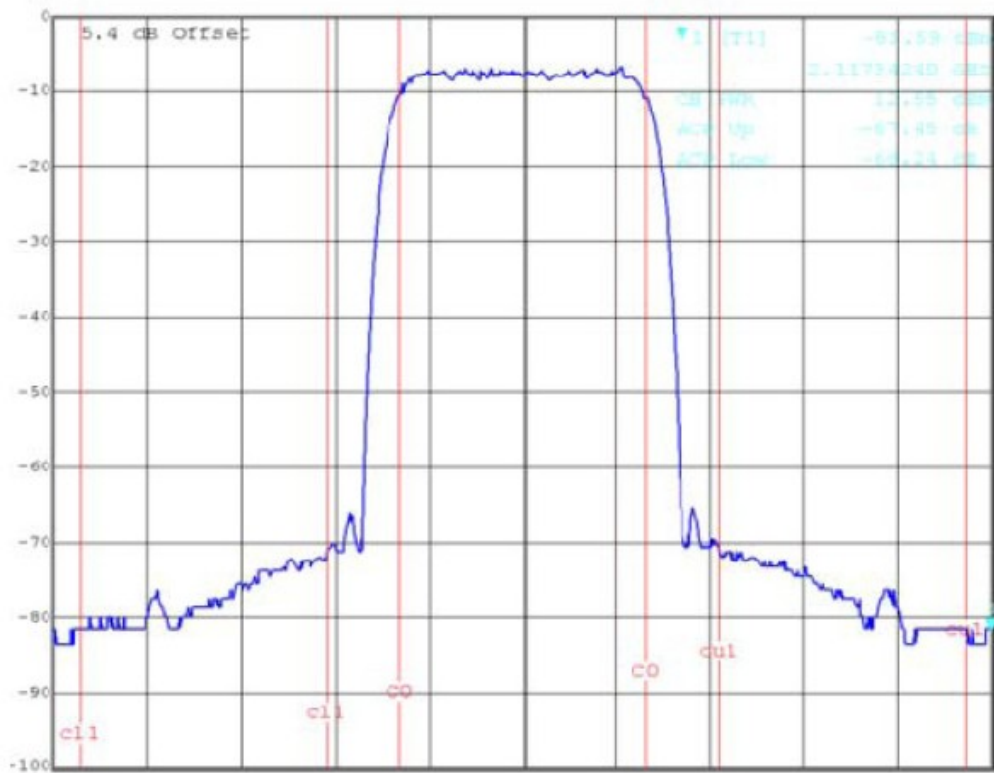
9.27

Nonlinearity: Mixer pyramid



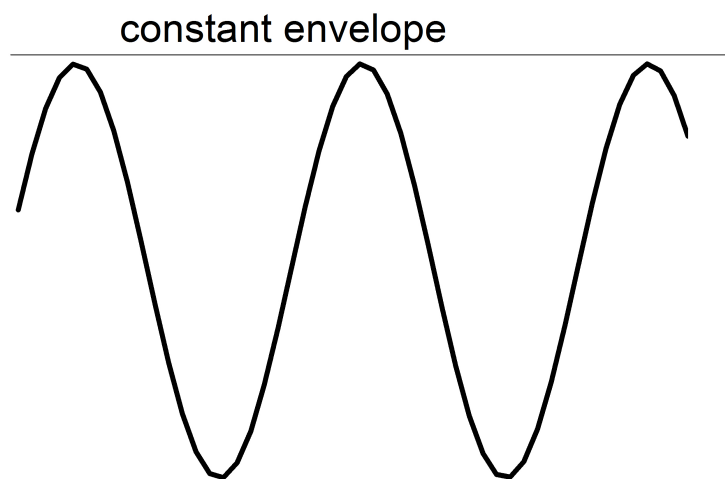
9.28

Modulated Signal Flows

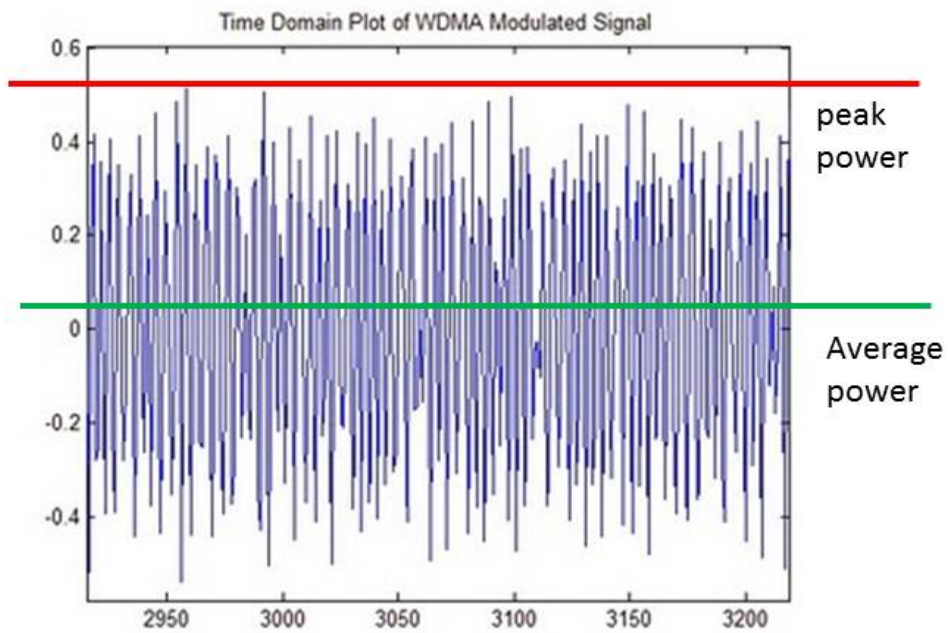


9.29

Constant Envelope Signals: Sine wave



9.30



9.31

3 Design for Linearity

22. june 2026

10.32

4 6G: 2030 Design for Linearity

IEEE Microwave Theory and Techniques Society

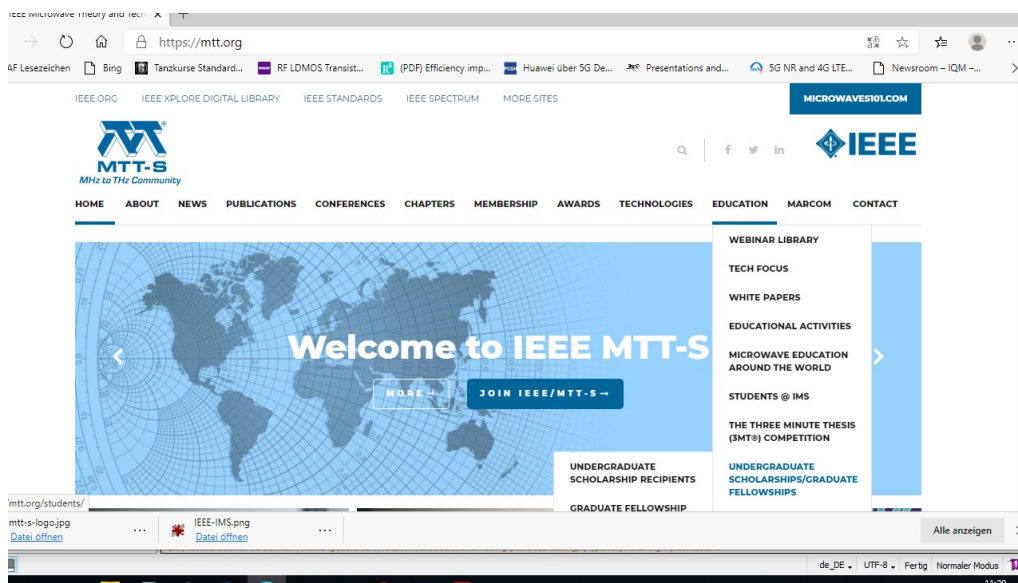
Challenges, scholarships, and money



10.33

IEEE Microwave Theory and Techniques Society

Challenges, scholarships, and money



10.34

IEEE Major Microwave Conference

International



Today

Motivation



10.36

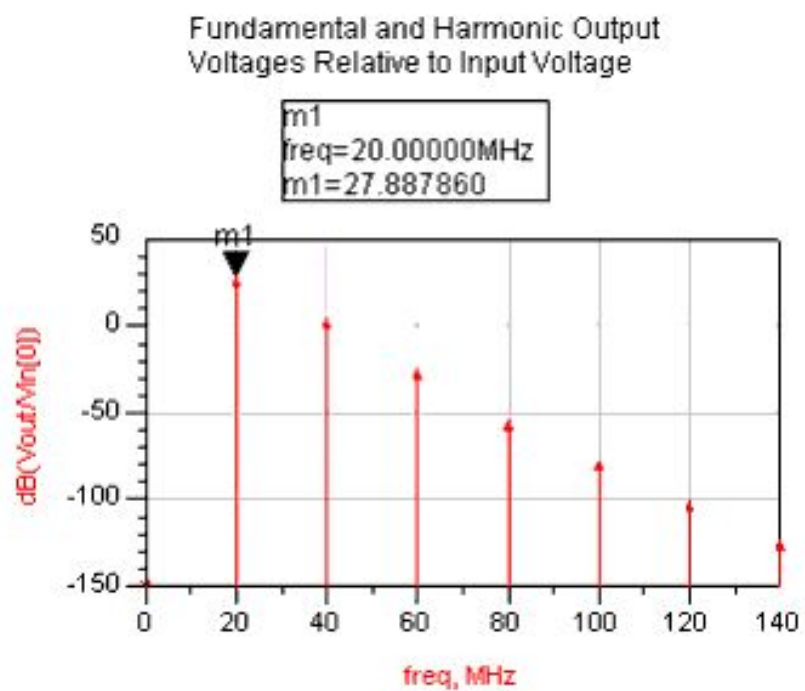
Efficiency vs. Linearity vs. Frequency

Real Base Station Demo

4.1 The classical trade-offs

Efficiency vs. Linearity vs. Frequency

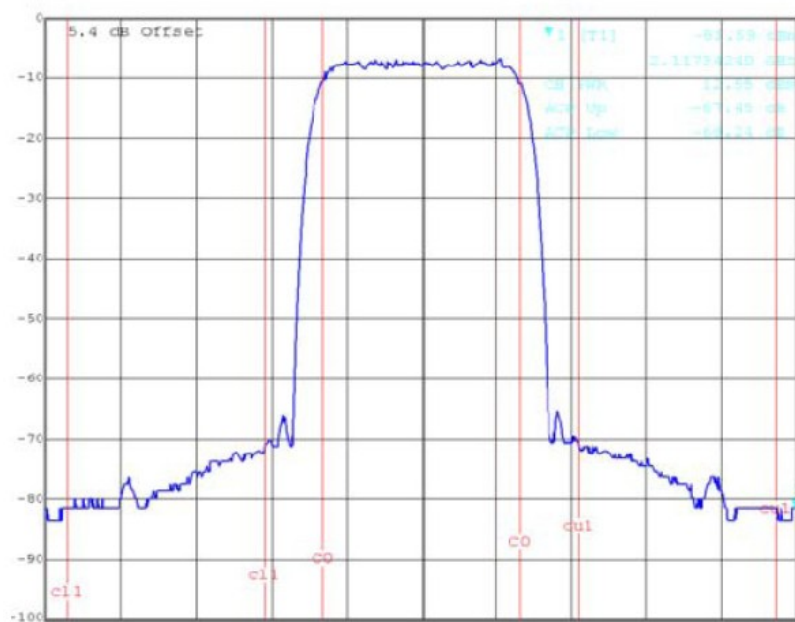
Simple: Harmonics vs. PAE



10.38

Efficiency vs. Linearity

Advanced: ACPR vs. PAE

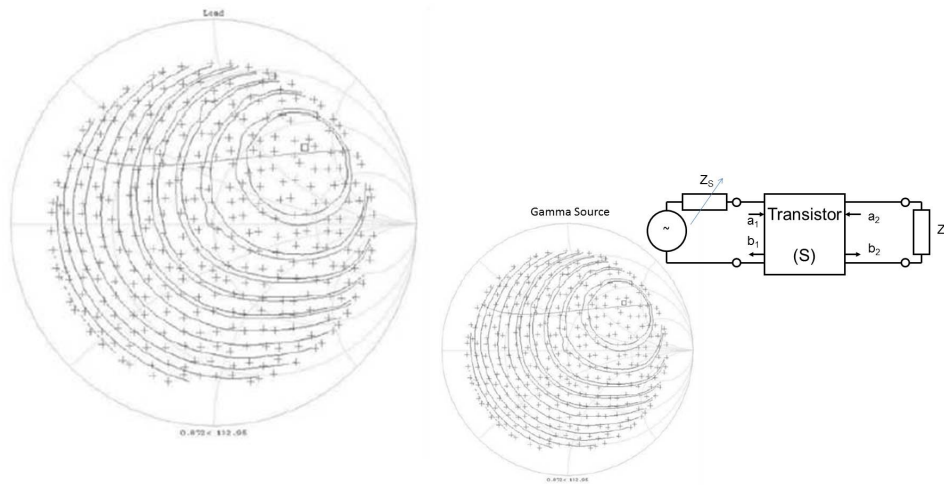


10.39

4.2 Hints: Factors of Influence to ACPR and PAE

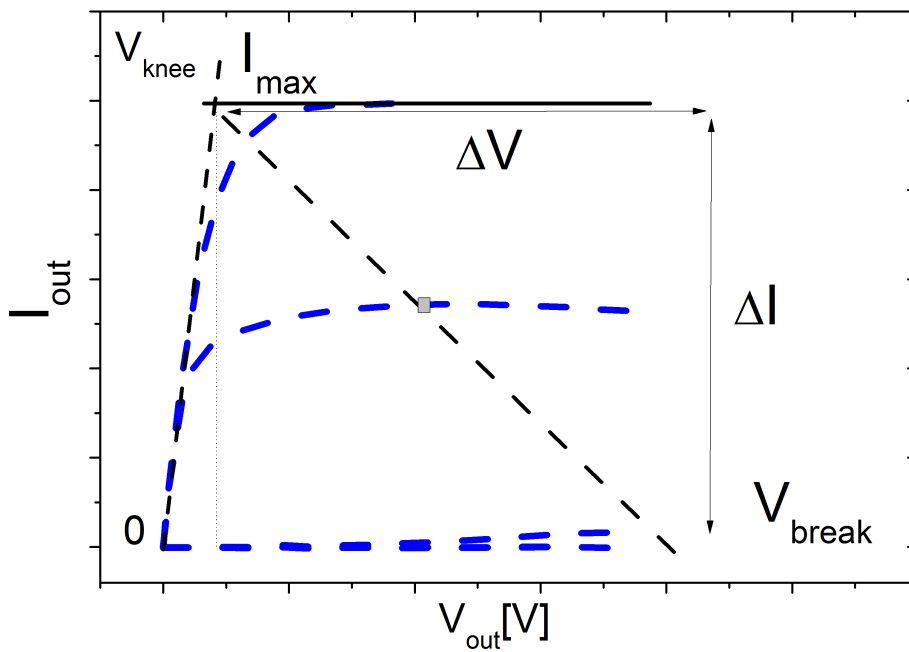
Fundamental Source and Load Impedance

Source and Loadpull



10.40

Bias: input and output bias !

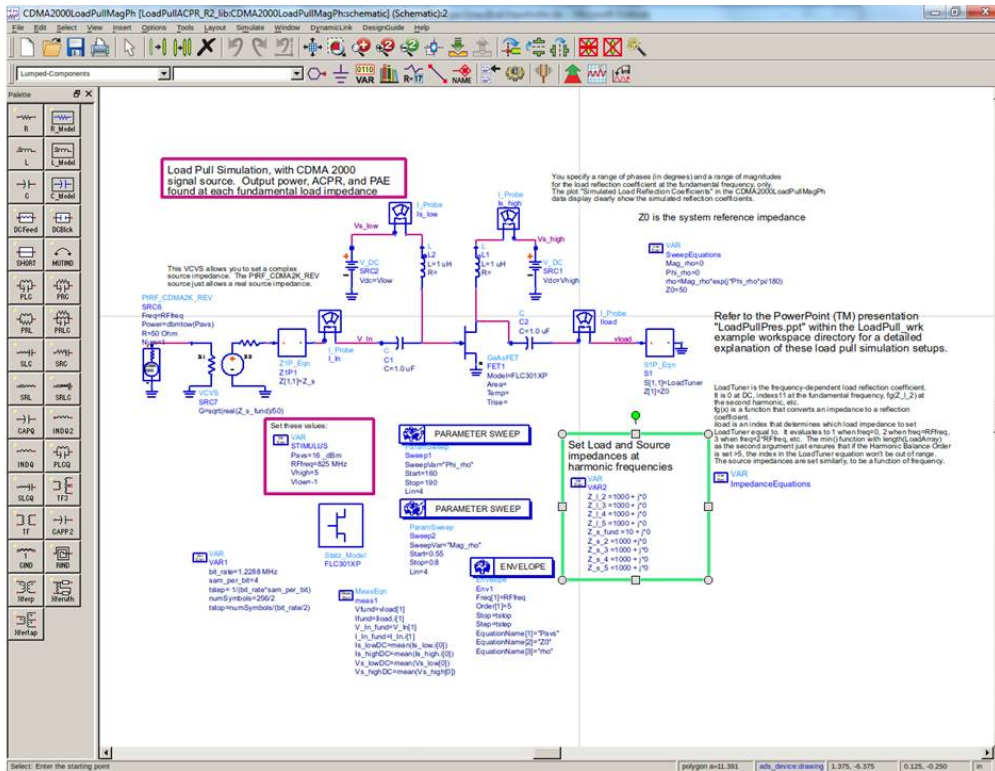


Be careful: the model might guide you to forbidden ranges of the bias.

10.41

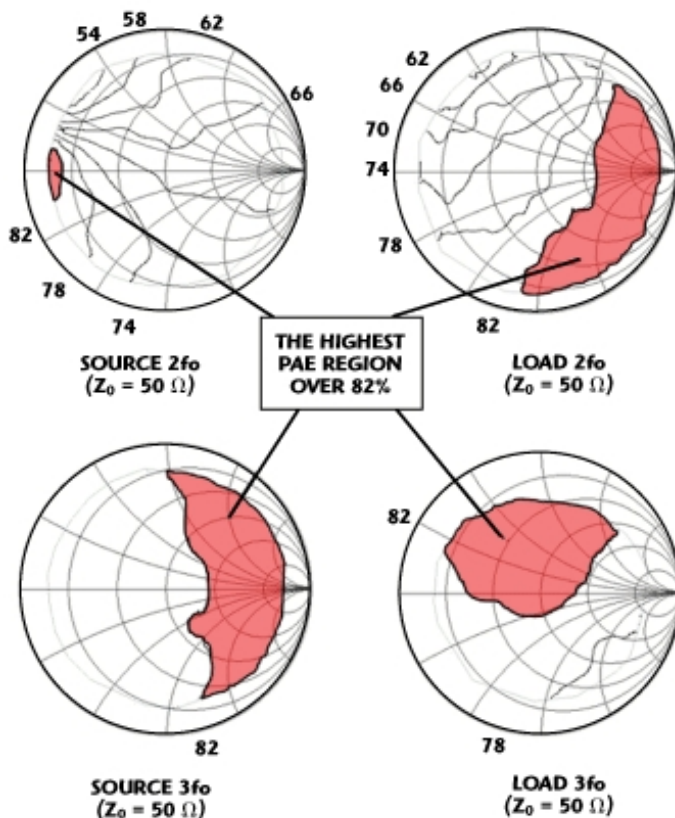
Potential is in: New and advanced: harmonics at input and output !

Reduce power in the harmonics through matching



Potential is in: Harmonics !

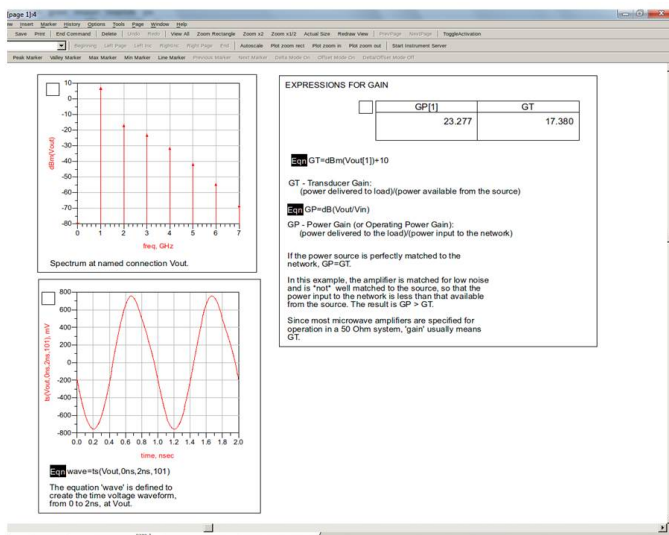
Harmonics !



10.43

Design Parameter Harmonics !

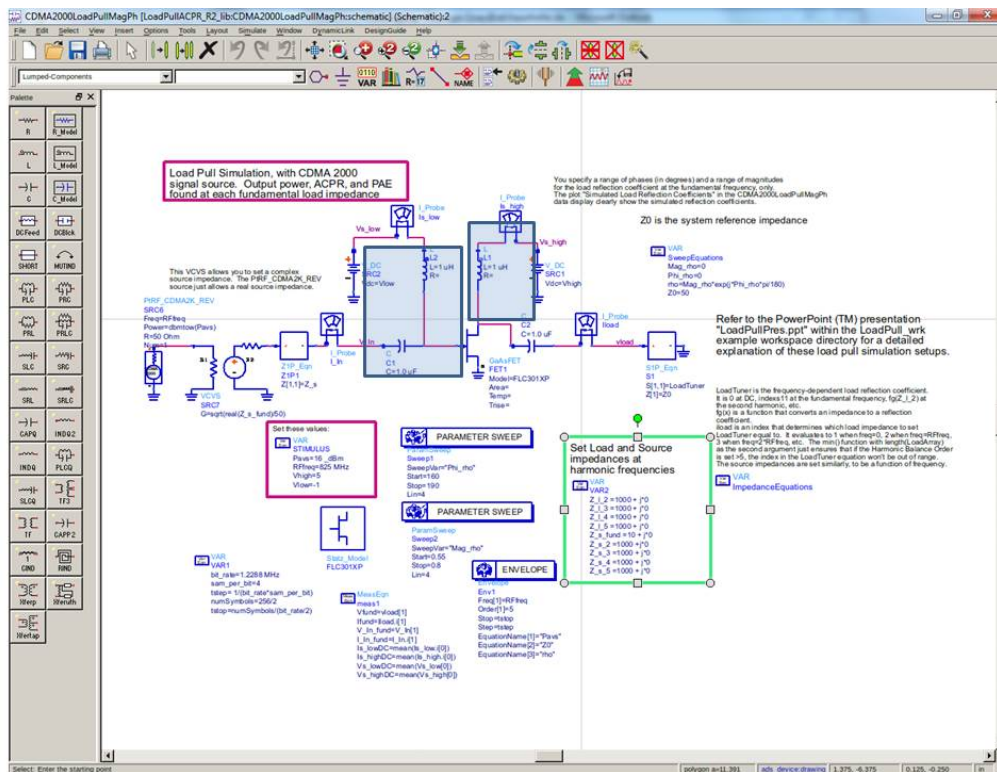
Harmonics !



10.44

Potential is in: New and Advanced: Baseband impedance !

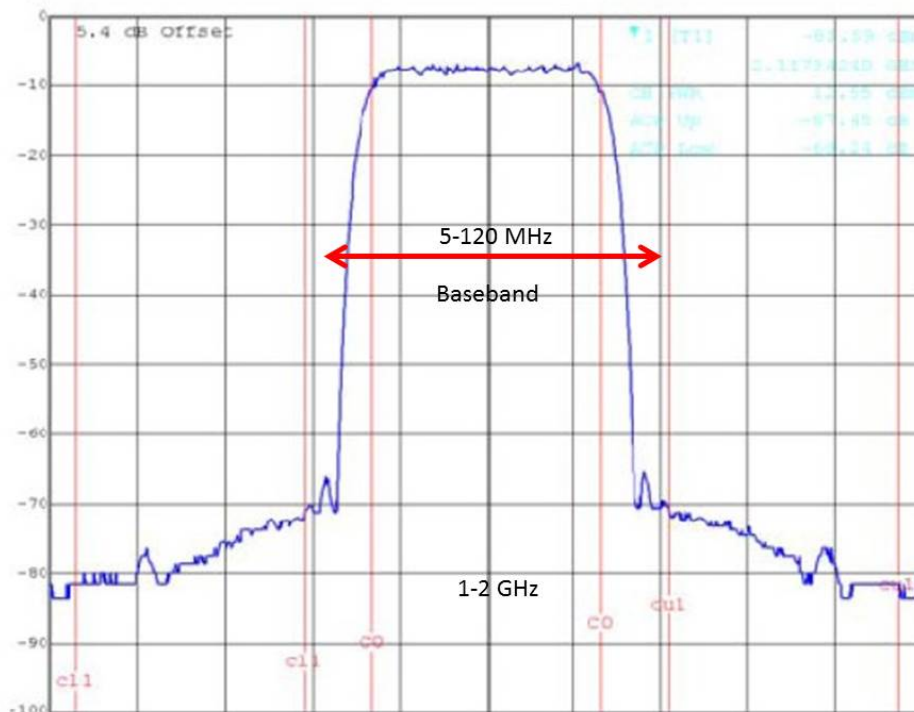
DC biasing defines an impedance for low-frequencies (MHz) !



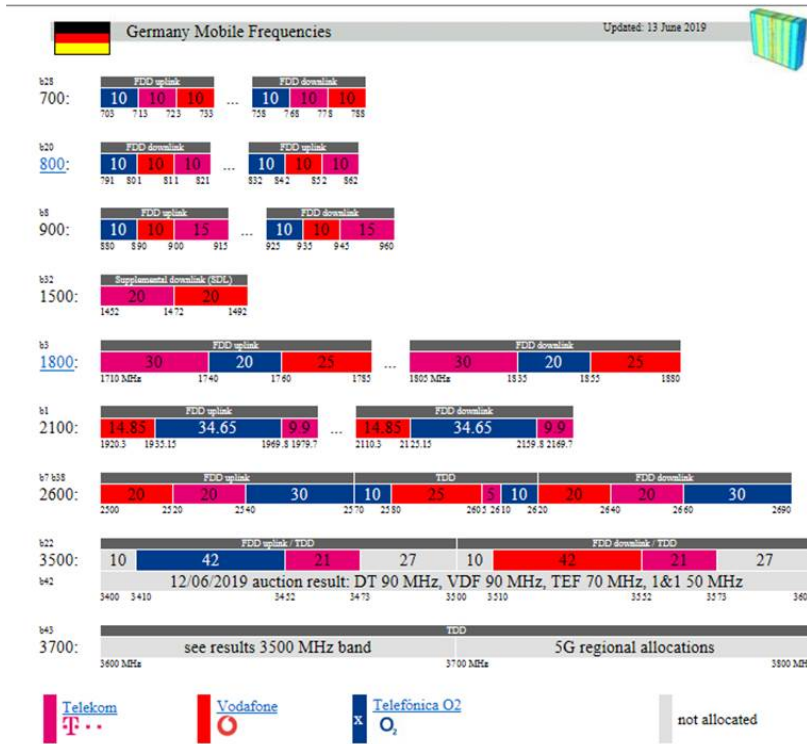
10.45

Efficiency vs. Linearity

Advanced: Baseband Impedance



10.46



10.47

Challenge

Optimize PAE under Constraints using many techniques

10.48

Challenge II

Winning team will be announced publically on the homepage

Presentation of results on homepage

A prize for the winning team: voucher of Amazon of 150 Euro

10.49

4.3 Tasks and Exercises: Vodafone Challenge

Tasks 16 block 4 part 2: Select challenge

1. Open examplesKCexamplesLoadpullACPR_R2_wrk
2. You are Telekom Germany, and, after buying frequencies, you have to build one amplifier, which converts two bands you just bought.
3. The challenge example will again do an analysis with a complex modulated signal for broadband operation. This signal will be the statistical signal used for 3G UMTS or CDMA (code division multiple access). We do not mix the standards, all is UMTS.
4. Challenge: Find one unique load setting, which at the same time maximizes the PAE at 700 and 900 MHz. We assume that this setting is frequency-independent.
5. You can change the number of simulated load state points as you like.
6. The challenge to you is to find the maximum average figure of merit: $PAE \times \text{frequency}$, taken for the two frequencies and two power levels, for a signal, which delivers
7. a.) an ACPR (down/up) of both -45 dBc or less for both ACPR (up and low) at the OUTPUT (not the input) for all frequencies.

10.50

Tasks 16 block 4 part 2: challenge

1. b.) Repeat the same procedure for an ACPR of -38 dBc and average the overall numbers for both frequencies and two power settings.
2. You may not change the model and the general set up of the exercise. The input power may be varied for each frequency, but the linearity must be held !
3. You can play with bias ($V_{DS} = V_{high}$ (3...6 V), $V_{GS} = V_{low}$ -3...0 V), fundamental input power, and the tuning of the fundamentals, if you like, the baseband matching of the amplifier. Bias must be the same of all frequencies.
4. This may also include the matching at harmonic frequencies at input and output and the changes of the so called baseband impedance, which you, in this case influence with the decoupling capacitances in the MHz regime. The same setting has to be valid for all two frequencies. One setting for the base band must be used.
5. A minimum output power of 20 dBm must be reached for all frequencies and the four power levels !

10.51

Tasks 16 block 4 part 2: challenge

1. The winner will be the PA that demonstrates the highest frequency-weighted power-added efficiency (PAE in percent, f in MHz), when producing an ACPR of 45 dB and 38 dBc, i.e. $((\sum_i PAE(45dBc)_i \times f_i)/2 + (\sum_i PAE(38dBc)_i \times f_i)/2)/2$.

10.52